**MR LUYOLO MAHAMBA**
071 006 6573 | luyolomahamba7@gmail.com | Johannesburg, Gauteng

## CAREER OBJECTIVE

I am a BSc student in Mathematics and Chemistry at the University of Johannesburg, passionate about applying quantitative reasoning, structured problem-solving, and logical thinking in software development. I have a growing interest in software engineering principles, object-oriented programming, database management, and user interface design.

My programming background in Java, SQL, and UI development complements my analytical mindset, preparing me to contribute to designing, developing, and maintaining robust software applications. I aim to grow professionally and personally through entry-level roles, graduate programs, and internship opportunities, while contributing with commitment, curiosity, and teamwork.

## EDUCATION

**Senior Certificate (Matric)**
School: Sithukuthezi Secondary School

**Bachelor of Science in Chemistry and Mathematics**
University: University of Johannesburg Duration: 2023
– Expected completion 2026 Key Modules:

- Calculus I & II (Rates of change, optimization, integrals)
- Linear Algebra (Vector spaces, matrices, and systems of equations)
- Differential Equations (Mathematical modelling and forecasting)
- Inorganic & Physical Chemistry (Critical reasoning and system analysis)
- Organic Chemistry
- Introduction to Algorithm Development (C++)
- Introduction to Data Structures (C++)
- Object-Oriented Programming

- Data Communications

## PROJECTS

**Inventory Management System (IMS) – Lead Developer**
Developed a JavaFX desktop application to help small businesses manage products, suppliers, and transactions.

- Designed intuitive UI with Scene Builder
- Built core features for managing inventory, transaction logging, and supplier tracking
- Integrated a MySQL database with JDBC, ensuring data integrity and persistence
- Implemented robust error handling and input validation

**Tech Stack**: Java, JavaFX, JDBC, MySQL, Scene Builder
GitHub: [github.com/Luyo200/Inventory-Management-System](github.com/Luyo200/Inventory-Management-System)

## SKILLS

**Mathematical & Analytical Skills**

- Strong foundation in functions, calculus, and systems modelling
- Analytical problem-solving using algebraic and logical reasoning
- Confident interpreting mathematical structures and trends in data
- Solid grasp of physics concepts to support quantitative analysis

**Programming & Technology**

- **Languages**: Java, C++, JavaScript, SQL, HTML, CSS
- **Tools**: GitHub, Eclipse, VS Code, Scene Builder, Notepad++
- **Databases**: MySQL (Basic relational design and queries)

- **Microsoft Office**: Word, Excel, PowerPoint

**Soft Skills**

- Fluent in English, IsiZulu, and IsiXhosa
- Detail-oriented, dependable, and team-focused
- Eager to learn and adapt quickly in dynamic environments

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**INTERESTS**

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- Designing and developing applications across platforms, including mobile apps, web applications, and desktop software
- Implementing mathematical models and algorithms to enhance app performance and functionality
- Creating simulations, automation tools, and utilities to support seamless user experiences
- Leveraging digital transformation and process automation to streamline development workflows
- Applying logic-based problem-solving to build scalable, user-friendly, and robust software solutions